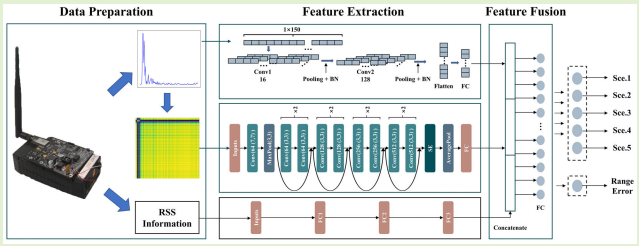
UWB NLOS Identification and Mitigation Based on Gramian Angular Field and Parallel Deep Learning Model

Bowen Deng, *Graduate Student Member, IEEE*, Tangwen Xu, and Maode Yan

***Abstract*—Ultrawideband (UWB) wireless localization technology has been widely applied in the field of indoor localization due to its good ability of noise resistance, strong penetration, and high measurement accuracy. However, the performance of UWB-based localization technology becomes poor when suffering from nonline-of-sight (NLOS) propagation conditions. Thus, it is necessary to identify NLOS propagation and mitigate the NLOS error. In this article, a novel NLOS identification and mitigation method based on multiinputs parallel deep learning model and Gramian angular field (GAF) is proposed. We utilize GAF to transform 1-D channel impulse response (CIR) signal into 2-D colored images, which adds additional high-level abstract features to the CIR signals. In the model training phase, the convolutional neural network (CNN) is used to extract temporal features from original CIR signals, and the residual network (ResNet) is used to extract visual features from GAF-encoded images. Besides, the received signal strength (RSS) information is also considered as an auxiliary feature to assist in identifying some NLOS scenarios with similar CIR features and further reduce the NLOS error. The experimental results show that our method has good ability in both line of sight (LOS) and NLOS binary classification and NLOS multiclassification, with accuracy over 96%. Additionally, based on the identification results, the proposed method can reduce the mean absolute error (MAE) and root mean square error (RMSE) of the range error from 65.61 and 96.82 cm to 4.19 and 6.95 cm, respectively. In the real indoor localization experiment, the proposed method can improve the localization accuracy by over 80%.**



***Index Terms*—Channel impulse response (CIR), Gramian angular field (GAF), multiinput learning, nonline-of-sight (NLOS) identification, ranging mitigation, ultrawideband (UWB).**

I. Introduction

HIGH-ACCURACY localization technology plays a more and more important role in the field of Internet-of-Thing (IoT). Different from outdoor environments, the mature localization technologies based on satellite systems such as Beidou and Galileo cannot provide real-time and high-accuracy localization services in indoor environments because of the poor penetration of the localization signal [1]. Thus, using wireless localization technologies such as radio frequency identification (RFID), ZigBee, long range radio (LoRa) and ultrawideband (UWB) to realize high accuracy indoor localization has become a hot research topic recently [2], [3], [4], [5]. However, not all of the indoor positioning technologies can meet the increasingly complex indoor scenarios. For instance, though LoRa can achieve long-distance localization it would not be suitable for large-scale commercial use because of its poor security. For the ZigBee positioning technology, it can be only used in some specific scenarios which do not request high real-time performance such as attendance system. Besides, the performance of the ZigBee positioning system is seriously affected by environmental factors, which means it cannot be used in harsh indoor environments. The RFID positioning technology can be divided into active RFID-based and passive RIFD-based. The former one is more susceptible to the environment and the latter is limited by its valid working distance, only a few meters. Consequently, the positioning technology mentioned above cannot meet the requirement of high-precision indoor positioning especially when the environment is complex. Among common wireless

Manuscript received 5 August 2023; revised 2 October 2023; accepted 7 October 2023. Date of publication 16 October 2023; date of current version 14 November 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFA1000303, in part by the National Natural Science Foundation of China under Grant 52372406, in part by the China Postdoctoral Science Foundation under Grant 2021M692740, in part by the Shaanxi Key Research and Development Plan under Grant 2022GY-297 and Grant 2022GY-255, in part by the Natural Science Foundation of Shaanxi Province under Grant 2021JQ-291 and Grant 2021JM-186, and in part by the Fundamental Research Funds for the Central Universities under Grant 300102328112. An earlier version of this article was presented at the Chinese Control Conference [DOI: 10.23919CCC58697.2023.10240284]. The associate editor coordinating the review of this article and approving it for publication was Dr. Vinay C. Gogineni. *(Corresponding author: Tangwen Xu.)*

The authors are with the School of Electronic and Control Engineering, Chang'an University, Xi'an 710064, China (e-mail: twxu@chd.edu.cn).

Digital Object Identifier 10.1109/JSEN.2023.3323564

1558-1748 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

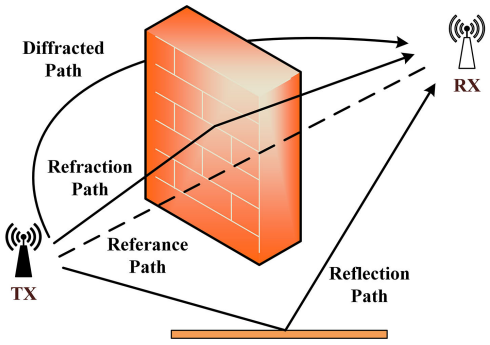


Fig. 1. NLOS propagation in the UWB-based localization system.

positioning technologies, UWB is considered the most reliable and promising technology which has already been widely used in industrial and challenging indoor environments such as underground mine and large-scale warehouse due to its admirable accuracy in long-distance positioning and good penetration [6]. In spite of the mentioned benefits, the performance of UWB experiences degradation when suffering from nonline-of-sight (NLOS) conditions.

At present, the signal NLOS propagation is the main factor that determines the accuracy of wireless indoor localization system [7]. The NLOS often occurs when the signal transmission path between anchor and tag is blocked by wall, goods, glass window, and other common indoor obstacles. Recent studies have shown that the main factor that NLOS affects the positioning result is that NLOS propagation will prolong the propagation time of ranging signals, which will lead to the poor performance of ranging-based location algorithms such as time of arrival (TOA) and time difference of arrival (TDOA) [8]. Fig. 1 shows the NLOS propagation in the UWB-based localization system. Moreover, the NLOS especially impacts the UWB positioning system work because the operational range of UWB is often over 100 m and the required positioning accuracy is in a few decimeters. Therefore, NLOS identification and mitigation are essential to enhance the performance of the UWB positioning system. Nowadays, many methods have been proposed to identify and mitigate NLOS propagation. However, most of them are subject to prior knowledge and require much human labor to extract useful information from the original channel impulse response (CIR) signal, which cannot meet the practical requirement. Besides, most of the existing methods only distinguish the line of sight (LOS) and NLOS propagation conditions, without considering the subdivision of NLOS scenarios according to certain obstacles causing NLOS, which cannot meet the increasingly complex environment, either. To deal with the above problems, in this article, we propose a novel NLOS identification and mitigation method based on Gramian angular fields (GAFs) encoding images and multiinput parallel learning (MIPL) network. The main contributions of this article are summarized as follows.

1) A novel and effective time-serial signal processing technology called GAF is applied in LOS/NLOS identification. In this article, we utilize GAF to transform CIR signal into 2-D image for NLOS identification and mitigation. Using GAF-encoded CIR images can solve the problem of feature loss when training convolutional neural network (CNN) directly using 1-D CIR signals. Moreover, it can also enable CNN to learn the high-level abstract features of the CIR signal, which can further improve the identification accuracy and reduce NLOS errors.
2) An MIPL model is designed and used. This model is composed of a CNN-based temporal feature extraction (TFE) module, a residual network (ResNet)-based visual feature extraction (VFE) module, and an auxiliary module. By using a parallel structure, we can extract features separately from heterogeneous data and subsequently achieve multifeature fusion. This approach not only improves the efficiency and stability of the model but also alleviates the over-fitting problem. To prove the advantages of the proposed method, several other NLOS identification and mitigation methods based on deep learning algorithms are evaluated.
3) Different from traditional LOS/NLOS binary identification, in this article, the NLOS scenario is subdivided based on common indoor obstacles, and the proposed method is tested in a real indoor multiscenario environment. The experimental results support that with the subdivision of the NLOS scenario, the NLOS mitigation performance of the proposed method can be further improved.
4) The received signal strength (RSS) features of UWB sensors in indoor LOS and NLOS scenarios are utilized to enhance the identification performance of subdivided NLOS scenarios with similar CIR features, thereby further mitigating the NLOS effect.

The rest of this article is organized as follows. In Section II, the previous works about NLOS identification and mitigation for UWB systems are briefly introduced. In Section III, the background and preliminaries of UWB NLOS identification and mitigation are explained. The details of the proposed UWB NLOS identification and mitigation method using parallel deep learning method are described in Section IV. The experimental details, results, and a brief analysis are presented in Section V. Some conclusions are given in Section VI.

II. Related Works

The existing LOS/NLOS identification and mitigation methods can be divided into two categories based on whether the machine learning method is used. In the following part, we will briefly introduce these two kinds of categories.

The traditional way to identify NLOS propagation does not use the machine learning method but focuses on analyzing the features of static parameters extracted manually from UWB CIR signals. Yu et al. [9] first extract static parameters like kurtosis and maximum amplitude from CIR signals. Then they use the Pearson correlation coefficient (PCC) method to select the optimal feature combination and use fuzzy comprehensive evaluation (FCE) to achieve NLOS identification. The hypothesis test is frequently used in NLOS identification. Gazzah et al. [10] analyzed RSS and angle of arrival (AOA) information by using the proposed hybrid hypothesis test to identify LOS/NLOS. Wang et al. [11] and Dong et al. [12] analyzed the features of RSS and angular

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

information to identify NLOS propagation. Some researchers focus on analyzing the manually extracted features from UWB signals using Gaussian methods [13], [14]. Silva and Hancke [15] investigated the statistical features of distance residuals including sum of the squares of distance residuals (SSDR), mean, standard deviation, and maximum, and they summarized a threshold to distinguish LOS and NLOS. Besides, some researchers constructed the probability density function (pdf) of range error to identify NLOS [16], [17]. Some researchers also identify NLOS by using extra environment information [18]. There are also some researchers who identify and mitigate NLOS by using additional equipment such as inertial measurement unit (IMU) sensors [19]. However, the traditional NLOS identification methods are always limited by prior knowledge. Besides, the static parameters should be extracted from UWB signals manually, which increases the difficulty of data preprocessing and sharply decreases the work efficiency. Moreover, the summarized threshold from static data may not be suitable for dynamic scenarios.

In recent years, machine learning theory has found extensive application in the analysis of complex data, which provides a novel method for NLOS identification. To alleviate the workload and improve the identification accuracy, traditional machine learning methods such as support vector machine (SVM) and random forest (RF) are applied in NLOS identification [20], [21]. Huang et al. [22] combined angular information with CIR signal and used traditional machine learning methods to identify NLOS. Similarly, Xiao et al. [23] investigated the correlation between RSS and NLOS, and used the least square SVM classifier (LS-SVMC) to classify LOS and NLOS. Moreover, some researchers combined semisupervised and unsupervised theory with SVM, and identified NLOS with labeled and unlabeled data [24], [25]. Compared with traditional NLOS identification methods, machine learning methods improve the LOS/NLOS classification accuracy and efficiency apparently. Nevertheless, traditional machine learning methods have high requirements for data processing. Besides, these methods still need to select vector features manually from the CIR time series, which costs much labor and cannot make full use of CIR signals.

Nowadays, more and more researchers focus on using original CIR signals to identify NLOS based on deep learning methods. Stahlke et al. [26] indicate that SVM cannot fully harness the features within CIR signals, resulting in lower identification accuracy compared to CNNs. In [27], the long short-term memory (LSTM) is used to analyze the temporal features of CIR signals. Additionally, in [28] the CNN and the LSTM are used successively to extract features from CIR signals for NLOS identification. To reduce the impact of background noises on NLOS identification, Jiang et al. [29] proposed a reversible transformation method to denosing the CIR signals and then used CNN to identify NLOS. Some researchers also apply the attention mechanism in traditional CNN and LSTM models, which helps in reducing interference from irrelevant data within the CIR signals [30]. However, the models are susceptible to over-fitting especially when datasets are small due to the use of LSTM. Pei et al. [31] introduced a self-attention mechanism into a fully CNN and implemented LOS/NLOS binary classification. In [32] and [33], researchers utilized continuous wavelet transform (CWT) to transform CIR time series into 2-D image, and then they investigated the difference between LOS and NLOS from the aspect of time-frequency features of CIR signals. Despite the promising accuracy of these methods, it is hard to select a proper wavelet basis to preprocess CIR signals collected in different scenarios. At present, the GAF has been used in the industrial field to preprocess vibration signals for mechanical fault diagnosis but has not been used in NLOS identification and mitigation [34], [35], [36].

In summary, the majority of existing NLOS identification and mitigation methods only consider LOS and NLOS scenarios but seldom consider subdividing the NLOS scenarios. Besides, these methods still rely on prior knowledge, and cannot fully explore the potential of original CIR signals on NLOS identification and mitigation. Moreover, most of the deep learning methods treat NLOS identification and mitigation as two separate problems, and often ignoring the potential to utilize identification results to enhance NLOS mitigation. As a result, these methods may not be suitable for practical application. Taking all factors into consideration, in this article, we propose a novel two-step NLOS identification and mitigation method based on GAF-encoded images and an MIPL model, which can make full use of the UWB CIR signal and require no prior knowledge. Besides, our approach can accurately identify multiple indoor NLOS scenarios, and the identification results are applied in NLOS mitigation to further enhance the dynamic positioning results.

III. Background and Preliminaries

A. UWB-Based CIR Signal in LOS/NLOS Condition

CIR represents how a channel responds to a short impulsive signal, which can describe the characteristics of a communication channel. For a UWB-based positioning system, the data transmission between anchor and tag can be expressed by [14]

$$s(t) = \sqrt{A} \sum_{m=0}^{M-1} p(t - mT_p) \tag{1}$$

where A is the amplitude of a single pulse, p is the pulse waveform, M is the total number of pulses that constitute an information frame, T_p is the sampling interval. The UWB channel impulse model can be modified as

$$h(t) = \sum_{i=1}^{L} a_i \delta(t - \tau_i) \tag{2}$$

where a_i and τ_i is the gain and time delay of path i, respectively, which depends on the signal propagation environments. The signal received by the UWB sensor is the accumulation of a single pulse, which can be expressed as

$$r(t) = \sum_{i=1}^{L} a_i s(t - \tau_i) + n(t) \tag{3}$$

where $n(t)$ is the additive Gaussian white noise. The CIR signals used in this article is extracted from the received waveform $r(t)$.

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

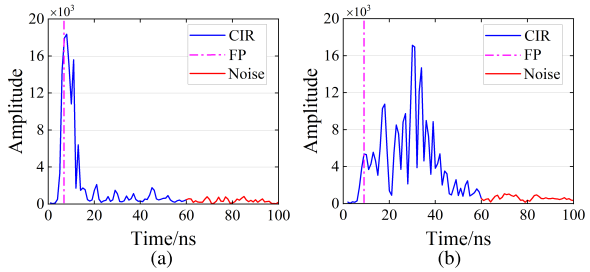


Fig. 2. UWB CIR signal under (a) LOS condition and (b) NLOS condition.

The normal operation of a UWB-based measurement system mainly depends on the signal transmission between anchor and tag. To measure the distance between the anchor and the tag, an anchor whose position is known will be necessary. Time of flight (TOF) is a frequently used method for distance measurement. Assuming that the result is only affected by NLOS, the measured distance between anchor and tag, using the TOF method, can be expressed by

$$\widetilde{d} = 0.5c[(t_d - t_a) - (t_c - t_b) + 2\tau_n] \tag{4}$$

where c is the signaling rate, t_a is the timestamp when the anchor first send CIR signal to the tag, t_b is the timestamp when the tag received the signal from anchor, t_c is the timestamp when the tag sent the response signal to the anchor, t_d is the timestamp when the anchor got response from the tag, τ_n is time-delay caused by NLOS.

From (4) we can know, compared with the LOS condition, that the ranging error caused by the NLOS condition always introduces a positive bias in the distance measurement. The CIR signal can be regarded as time series. Fig. 2 shows the CIR curves under the LOS condition and NLOS condition, from which we can see that the LOS CIR curve is different from the NLOS CIR curve. For the LOS CIR curve, its first path (FP) can always reach the peak, while the FP of the NLOS curve usually cannot reach the peak. Besides, the maximum value of the LOS CIR curve is also larger than that of the NLOS CIR curve. Moreover, we can see that the NLOS CIR curve is affected by the multipath effect, which results in multiple local peaks in the waveform. In short, the CIR signal has different features when transmitted from different scenarios, which makes it possible to distinguish LOS and NLOS by using of machine learning method.

B. Convolutional Neural Network

In recent years, CNN has been widely used in fault diagnosis. And with the development of attention mechanisms, the performance of CNN has been further improved. A typical CNN consists of three main layers, a convolutional layer, a pooling layer, and a fully connected (FC) layer.

1) Convolutional Layers: The convolutional layer is the core of CNN, which is used to extract features from original data. When the model input is 1-D data like a time series, to achieve feature extraction, the kernel filter whose size is far less than the input data will assign weights and bias to each parameter in the time series, this process can be modeled as [37]

$$y_i^{(l)} = a\left[\sum_{\gamma=1}^{N_k^{(l)}} \omega_r^{(l)} x_{r+i\times N_s^l} + b^{(l)}\right] \tag{5}$$

$$0 \le i \le \frac{N - N_K^{(l)}}{N_S^{(l)}} \tag{6}$$

where $N^{(l)_K}$ is the kernel size of filter, $\omega_r^{(l)}$ and $b^{(l)}$ are the weights and bias of the filter, respectively, $l = 1, 2, \ldots, L$.

At the end of the convolution operation, the outputs $Y = [y_i^1, y_i^2, \ldots, y_i^l]$ are fed into an activation function to introduce nonlinear features into neural networks, which improve the performance of the linear model in classification. In this article, the rectified linear units (ReLU) activation function is adopted as it can effectively alleviate model overfitting. ReLU can be expressed by

$$f_{\text{ReLu}}(x) = \begin{cases} x, & x > 0 \\ 0, & \text{else.} \end{cases} \tag{7}$$

2) Pooling Layers: In order to reduce the computation load and time complexity of the neural network, the pooling layers are introduced to further processing the outputs of convolutional layers by means of down-sampling. In this process, the key features of data are preserved whereas the rest are abandoned. In this article, the max pooling method is selected. The max pooling method, which has been proven to be effective in feature extraction, is the most frequently used method in recent years. During the operation, the data will be divided into several parts based on the kernel size. The max value in each part will be preserved and used to rebuild a new matrix that contains the critical features of the origin data.

3) Fully Connected Layers: FC layers are the last part of the CNN. The extracted features are fed into FC layers and flattened into one vector, during which a Softmax activation function that limits the outputs between 0 and 1 by weight is used to achieve multiclassification. The linear activation function is used to predict the range error. The Softmax and Linear activation function can be expressed by

$$\text{Softmax}(x) = \frac{e^{\omega_j^{(i)} x_j}}{\sum_j^n e^{\omega_j^{(i)} x_j}} \tag{8}$$

$$\text{Linear}(x) = x_j \tag{9}$$

where $X = [x_1, x_2, \ldots, x_i]$ is the feature vector of inputs, ω is the weight, $i \in (0, k)$ is the number of categories.

IV. Proposed Method

In this section, we first introduce the necessity of NLOS scenarios subdivision. Subsequently, a method that transforms CIR signals into images by using GAF is presented to explore the high-level visual feature of CIR signals. Last but not least, the structure and workflow of the designed MIPL model are described.

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

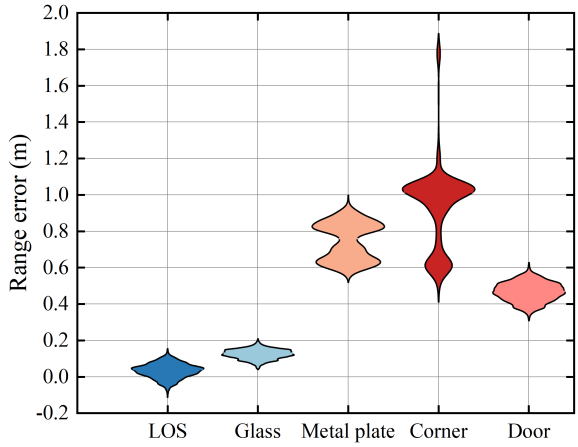


Fig. 3. Violin plot of ranging error under LOS and four NLOS propagation conditions.

A. Subdivision of NLOS Condition

The distribution of NLOS range errors caused by different indoor obstacles is irregular. The violin plot which contains the probability density information of range errors in LOS and four typical indoor NLOS scenarios is presented in Fig. 3. As shown, we can see that when UWB CIR signals transmit under LOS conditions, the range error is regular and varies within a small range. On the contrary, the overall range error is larger when under NLOS conditions. Among the four types of NLOS range error, the range error caused by corner is the largest and fluctuates in a large range. When it comes to the glass, the ranging errors are much less than others. In a word, the range error varies greatly when the obstacles are different. Therefore, when locating a dynamic UWB tag, the dynamic changes in NLOS scenarios can seriously affect the localization accuracy, leading to unsatisfactory performance of existing LOS/NLOS identification and mitigation methods that have not considered subdividing NLOS scenarios. All things considered, we subdivided indoor NLOS scenarios into multiple typical indoor scenarios, rather than LOS and NLOS two simple scenarios.

B. Gramian Encoding of CIR Signal

Time series reconstruction techniques have been widely used in various domains, particularly in machine learning. In this article, we propose a novel approach to convert CIR signals into 2-D images utilizing GAF. This approach preserves the essential features of CIR signals and takes full advantage of CNN.

The GAF encoding process consists of three steps. To begin with, the original CIR signals $X = [x_0, x_1, x_2, \ldots, x_n]$ should be normalized to between 0 and 1, which can be expressed by

$$\widetilde{x_i} = \frac{x_i - \min(X)}{\max(X) - \min(X)}, \quad i \in [0, n]. \tag{10}$$

Then the normalized signal $\widetilde{X} = [\widetilde{x_0}, \widetilde{x_1}, \widetilde{x_2}, \ldots, \widetilde{x_n}]$ is converted from Cartesian coordinate into polar coordinate, which can be expressed by

$$\begin{cases} \phi_i = \arccos(\widetilde{x_i}), & 0 \leqslant \widetilde{x_i} \leqslant 1 \\ r_i = \dfrac{t_i}{N} \end{cases} \tag{11}$$

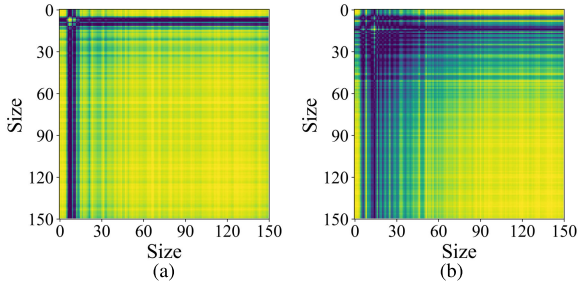


Fig. 4. GAF encoding CIR signals from typical (a) LOS and (b) NLOS.

where ϕ_i is the angle polar coordinates, r_i is the radius, t_i and N is the timestamp of x_i and the constant value, respectively, both of which determine the range of the time span in polar coordinate.

After the transformation to polar coordinate, the obtained parameters are fed into the Gramian angular summation field (GASF) function, and the function results form the Gramian matrix whose elements can be considered as pixels forming images. Consequently, the CIR signal is converted into a 2-D matrix. This operation can be modeled as

$$G_{ij} = \cos(\phi_i + \phi_j), \quad i, j \in [0, n] \tag{12}$$

$$\text{GASF} = \begin{bmatrix} G_{11} & \cdots & G_{1n} \\ \vdots & \ddots & \vdots \\ G_{n1} & \cdots & G_{nn} \end{bmatrix} \tag{13}$$

where G_{ij} is the GASF equation, n is the length of time series.

According to trigonometric transformation, (12) and (13) can also be expressed as

$$\text{GASF} = \widetilde{X}' \cdot \widetilde{X} - \sqrt{I - \widetilde{X}^2}' \cdot \sqrt{I - \widetilde{X}^2}. \tag{14}$$

Fig. 4(a) and (b) show the GAF transformation results of LOS and NLOS CIR signals. As shown, all data in the CIR signal is retained and transformed into visual domain. The dark lines represent the key data for NLOS identification and mitigation in the CIR signal, while the other parts represent the background noise. Apparently, in Fig. 4, the GAF-encoded LOS and NLOS CIR signals exhibit their individual features in terms of the visual domain.

C. Multiple Input Parallel Learning Neural Network

In this article, we propose an efficient MIPL model for classifying UWB CIR signals propagation conditions and mitigating the NLOS effect. Different from the traditional single input model, the MIPL model combines temporal and visual features of the CIR signal. Meanwhile, the distribution features of signal strength are also considered. In some cases, the features of signal strength are not apparent, but they become evident when the signal transmitting under harsh NLOS environments. By using the MIPL model, the features of heterogeneous data can be extracted in different algorithms and finally fused into composites, which improves the efficiency of the deep learning model and alleviates the over-fitting problems. The overall framework of the MIPL model is shown in Fig. 5. During the model training phase, the data with scenario labels and range labels will be fed into the MIPL model for classification and error prediction,

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

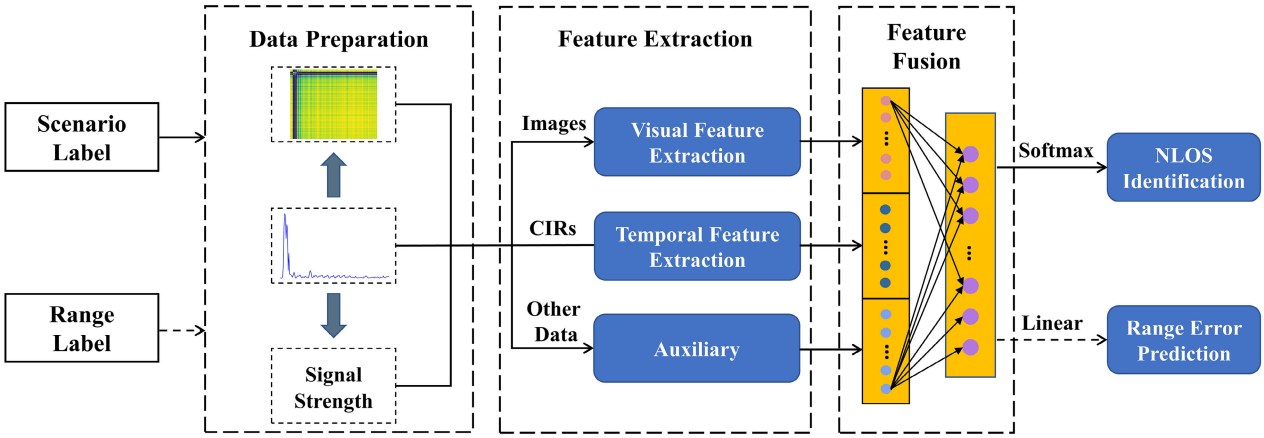


Fig. 5. Overall framework of multiple inputs parallel deep learning model.

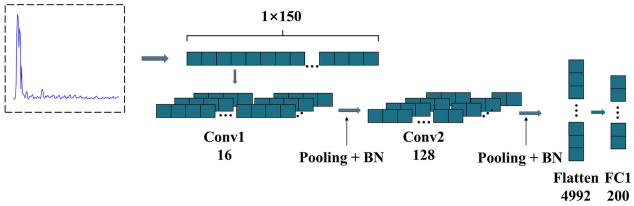


Fig. 6. TFE module.

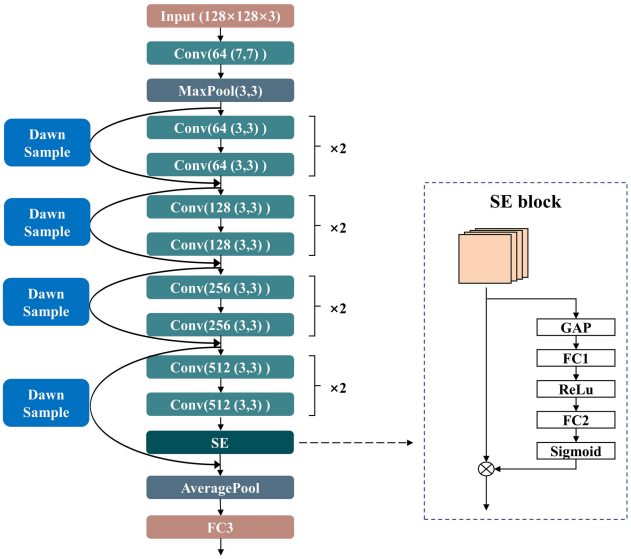


Fig. 7. VFE module.

respectively. As shown, the key steps of the MIPL model consist of four modules.

1) Temporal Feature Extraction Module: Recent studies have indicated that recurrent neural networks (RNNs) have good performance in data forecasting. Nevertheless, for the classification of time series data, CNN shows better performance, especially when data is long or datasets are small. Therefore, in this study, the CNN is used to extract temporal features from CIR signals. Fig. 6 shows the detailed structure of the TFE module. As shown, the TFE module consists of two 1-D convolutional layers. A max pooling layer and a batch normalization layer are added after each convolutional layer to address the vanishing gradient problem and enhance the speed of model convergence. At the end of the TFE module, a flatten layer and a dense layer are added, where the extracted features are transformed into a 1-D vector, and the feature size is reduced to balance the information from other modules.

2) Visual Feature Extraction Module: In order to make full use of the GAF-encoded images, in this study, we use ResNet to construct the VFE module. Fig. 7 shows the framework of the VFE module. The VFE module is consists of several residual blocks. Each residual block has different kernel sizes and input channels. Although GAF-encoded images can keep all the features in the origin CIR signals, the background noises are also included, which will affect system efficiency. Therefore, at the last residual block, a squeeze-and-excitation (SE) block is added to weaken the irrelevant features of GAF-encoded images. The last part of VFE module also consists of a flatten layer and a dense layer to rescale the extracted features.

3) Auxiliary Module: The pipeline of the auxiliary module is shown in Fig. 8. The main purpose of this module is to support the MIPL model in better achieving classification and prediction. The auxiliary module is composed of an input layer and three hidden layers. The other data such as RSS and angular information can be fed into this module after data splicing to extract extra features, which can further improve the efficiency and capability of the neural network to implement scenario classification. In this study, the signal strength information obtained directly from the UWB sensors, including RSS and the received power of the FP, is utilized to assist in both NLOS identification and mitigation processes.

4) Feature Fusion Module: The feature fusion module is the last part of the MIPL model. As shown in Fig. 5, the feature fusion module consists of a merge layer and a hidden layer. The rescaled feature vectors from three feature extraction modules are connected and fused into a new vector. Finally, the new feature vector is fed into a hidden layer with Softmax active function for classification and another hidden layer with a linear active function for error prediction, respectively.

Furthermore, during the model training phase, the cross-entropy loss function is used to assess whether the model is overfitting. Additionally, the learning rate is automatically

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

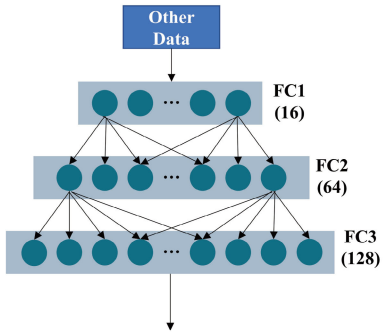


Fig. 8. Auxiliary module.

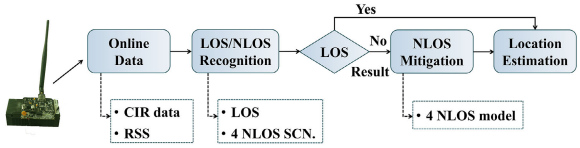


Fig. 9. Workflow of NLOS identification and mitigation.

adjusted based on the loss function results, which effectively addresses overfitting issues.

The workflow of the proposed MIPL method is shown in Fig. 9. In the beginning, the online data and the trained classification model are used to identify the propagation condition, which includes LOS and four subdivided NLOS scenarios. Then, if the identification result indicates LOS, the range information will be directly utilized for localization. Otherwise, the appropriate NLOS mitigation model will be selected to predict range error, ultimately mitigating the impact of NLOS.

V. Experimental Setup and Verification

In this section, the composition of the dataset is first introduced. Then, some frequently used performance metrics are chosen to evaluate the performance of NLOS classification results and error prediction results. After that, we test the binary classification performance and the multiclassification performance of MIPL, respectively. In both experiments, CNN, staked-LSTM, SVM and other popular classification methods are selected to be the comparison. Additionally, we test the accuracy and the performance of MIPL on error mitigation. Two traditional deep learning methods (CNN and LSTM) with optimal models are also selected to further demonstrate the robustness of the MIPL method. Moreover, we tested MIPL methods with undivided NLOS data and subdivided NLOS data, respectively, to verify whether NLOS scenarios subdivision would affect error mitigation results. Finally, the proposed method is tested in a real indoor environment.

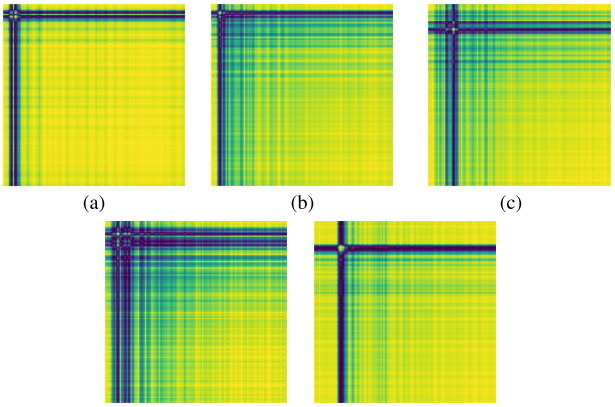
A. Dataset Description and Performance Metric

To verify the performance of the MIPL model in NLOS identification and mitigation, in this study, 4000 LOS CIR samples and 8500 NLOS CIR samples (2500 from glass window, 2500 from corner, 3000 from metal plate and 500 from door) are collected from office and corridor shown in Fig. 10(a) and (b). The data is collected by DW1000-based UWB devices with center frequency of 4.0 GHz, data rate of



(a) (b)

Fig. 10. Instruments and experimental environments. (a) Office. (b) Corridor.



(a) (b) (c) (d) (e)

Fig. 11. (a) GAF encoding LOS CIR curve, and NLOS CIR curve of four scenarios. (b) Glass window. (c) Metal plate. (d) Corner. (e) Door.

110 kb/s and pulse reception frequency of 64 MHz. Besides, all data measurements are based on the UWB IEEE 802.15.4a standard channel model. To reduce the computational complexity, each CIR signal only retains 150 essential parameters containing key information.

Before the measurement, the coordinates of each anchor are measured first. Then, during the measurement the tags are moved around in a space with some typical indoor obstacles to collect CIR signals with different features. At the same time, the E50 laser rangefinder with a range error of under 5 mm while the range less than 100 m, is utilized to measure the distance between tags and anchors as the true value. Next, the measured values of distance are obtained by calculating the CIR signal propagation time. After that, the ranging error can be obtained by comparing both values. To avoid the overfitting problem caused by some special positions, each type of data is mixed together randomly. In this article, we divided 12 500 samples into 70% training dataset and 30% test dataset. Fig. 11 shows GAF encoding images collected from the LOS scenario and four different NLOS scenarios.

To evaluate the model performance, in this work, we choose four frequently used performance metrics called accuracy, mac-precision, mac-recall, and mac-F1-score to evaluate classification results, and we use mean absolute error (MAE) and root mean square error (RMSE) to evaluate range error mitigation results. Accuracy is usually used in binary classification, which indicates the number of correctly classified samples accounts for the total number of samples, which can be expressed by

$$\mathrm{Acc} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{15}$$

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

Mac-precision, mac-recall and mac-F1 are used to measure model prediction accuracy in multiclassification, which can be defined as follows:

$$P_{\text{mac}} = \frac{1}{n}\sum_{n=1}^{n}\frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i} \tag{16}$$

$$R_{\text{mac}} = \frac{1}{n}\sum_{n=1}^{n}\frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i} \tag{17}$$

$$F_{\text{mac}} = 2 \times \sum_{n=1}^{n}\frac{P_{\text{mac}} \times R_{\text{mac}}}{P_{\text{mac}} + R_{\text{mac}}} \tag{18}$$

where $i = 1, 2, 3, 4, 5$ is the five types of CIR signal propagation channel, true positive (TP) is the correctly classified positive samples, true negative (TN) is the correctly classified negative samples, false positive (FP) is the misclassified positive samples and false negative (FN) is the misclassified negative samples.

The MAE and RMSE are used to evaluate the result of prediction. Usually, the less the MAE and RMSE are, the more accurate the prediction result would be, both of which are defined as follows:

$$\text{MAE} = \frac{1}{n}\sum_{n=1}^{n}\left(\left|d_i' - d_i\right|\right) \tag{19}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{n=1}^{n}\left(\left|d_i' - d_i\right|\right)^2} \tag{20}$$

where d_i' is the real value and d_i is the measured value of ranging, n is the number of data.

B. Experimental Setting Up and Model Evaluation

In this article, the proposed MIPL method is evaluated from four aspects. First, on the one hand, we evaluate the performance of the proposed method for traditional LOS/NLOS scenario binary classification. Second, the performance of LOS and multi-NLOS scenarios classification is also considered to further evaluate the proposed method. Third, we also test the proposed method of error mitigation. Finally, the proposed method is tested under real indoor localization experiments. All the experiments select other popular deep learning methods with optimal models to be the comparison. The experiments were conducted on a PC equipped with an AMD R7-5800H CPU, 16 GB of RAM, and an NVIDIA RTX 3060 graphics card (with 6 GB of video memory), running on a 64-bit Windows 10 operating system. Additionally, the software platform used was Python 3.8 with TensorFlow 2.5.

1) Performance Evaluation for Traditional LOS/NLOS Binary Classification: For the purpose of evaluating the performance of the MIPL model in traditional binary classification, the MIPL model is compared with CNN, stacked-LSTM, CNN-LSTM, SVM and ResNet five traditional methods. For the former four methods, their inputs are the normalized CIR signals, which is the common approach for LOS/NLOS classification. For the ResNet, its inputs are the GAF encoding images. Besides, we divide the MIPL method into MIPL-A and MIPL-B. The MIPL-A method is only trained with CIR signals with their corresponding GAF encoding images. whereas MIPL-B is also considered the RSS features.

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT METHODS FOR TRADITIONAL BINARY CLASSIFICATION

Method	Precision	Recall	Accuracy	F1
CNN	87.32%	91.51%	85.49%	89.36%
stacked-LSTM	80.84%	85.87%	77.04%	83.28%
CNN+LSTM	85.65%	88.28%	82.35%	86.95%
SVM (linear)	86.07%	87.55%	82.28%	86.81%
ResNet	88.36%	87.84%	84.20%	88.11%
MIPL-A	92.71%	94.43%	91.35%	93.56%
MIPL-B	98.55%	97.47%	97.36%	98.01%

Table I shows the LOS/NLOS binary classification performance comparison of different methods. Four performance metrics to evaluate each machine learning method are shown in Table I. It can be seen that the four metrics of both MIPL-A and MIPL-B are above 90%, and they can better identify the LOS scenario and NLOS scenario than other traditional methods. Compared to MIPL-A, the precision, recall, accuracy and F1 of MIPL-B are 5.84%, 3.04%, 6.01% and 4.45% higher, respectively, which indicates that signal strength information can improve the classification accuracy. Then, we can see that among five traditional methods, the accuracy of CNN (85.49%) is the highest compared to stacked-LSTM (77.04%), SVM (82.28%) and ResNet (84.20%). On the contrary, LSTM performs the worst among these methods. As mentioned before, LSTM-related methods are applicable to time series prediction. However, when it comes to LOS/NLOS classification, these two approaches cannot extract sufficient features from CIR signals with low time complexity. Moreover, the performance of the LSTM-related method is constrained by the size of the dataset. In this work, the dataset contains about 12 500 CIR data, which makes it rather difficult to control overfitting while ensuring accuracy. For the SVM, the lack of high-dimensional features of the CIR signal can account for the poor performance of the SVM. As for the ResNet, the accuracy of which is lower than CNN but higher than other traditional methods. Although GAF encoding image can bring visual features to the original CIR signal, its temporal feature cannot be extracted sufficiently and sometimes even lost during multiple convolution operations, which accounts for the lower accuracy of ResNet compared with CNN.

Additionally, the receiver operating characteristic (ROC) curves shown in Fig. 12 are presented to further evaluate the performance of the MIPL model. It is obvious that the area under corner (AUC) of MIPL-A (0.969) and MIPL-B (0.997) is higher than other traditional methods, which means that MIPL has a better performance in LOS/NLOS binary classification. In a word, the combination of temporal features and visual features can improve the accuracy of LOS/NLOS binary classification. However, whether MIPL method is suitable for more complex scenarios still needs further discussion.

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

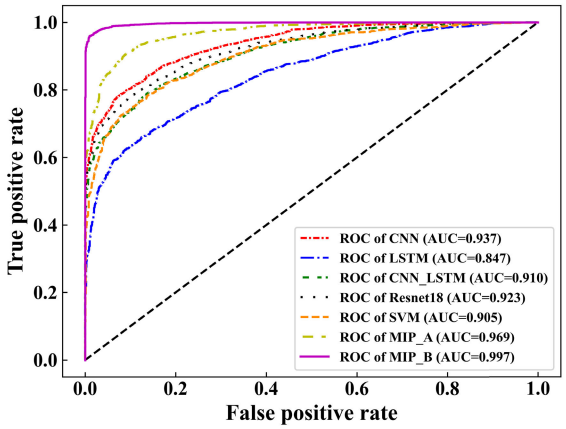


Fig. 12. ROC curve for different methods.

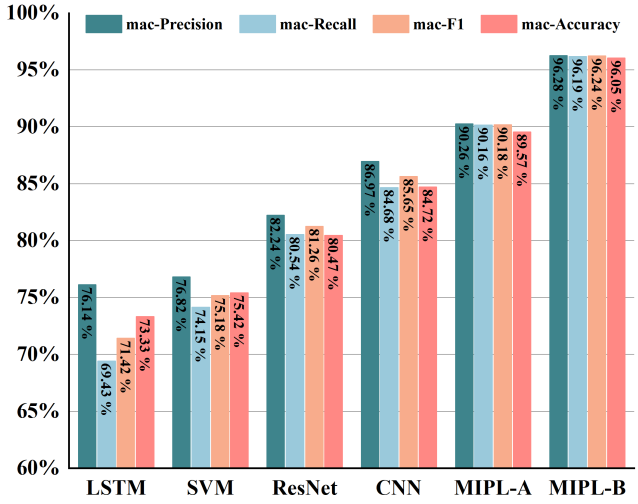


Fig. 13. Performance comparison of different methods for multiclassification.

2) Performance Analysis of Multiple Scenarios Classification: To further evaluate the performance of MIPL for complex scenario classification, we test the MIPL method on scenarios multiclassification. The NLOS scenario is subdivided into four common indoor NLOS scenarios including glass window, metal plate, corner, and door. In the following experiment, the MIPL method is also divided into MIPL-A and MIPL-B. Fig. 13 shows mac-precision, mac-recall, mac-F1 and accuracy of four traditional methods and two MIPL methods. The performance of traditional methods deteriorates when dealing with scenario multiclassification problems, especially for the LSTM-related methods and SVM, which indicates that the traditional methods cannot extract enough features to distinguish different scenarios. Obviously, the performance of the two MIPL methods in multiscenario classification is still better than that of the traditional methods. Compared to MIPL-A, the mac-precision, mac-recall, accuracy and mac-F1 of MIPL-B are 6.02%, 6.03%, 6.06%, and 6.48% higher, respectively. Different from MIPL-A, the MIPL-B considers the distribution features of signal strength which helps to distinguish scenarios with similar CIR features and thus improves the overall performance. To prove this, the principal components analysis (PCA) is used to show the feature distribution map (see Fig. 14) of each method.

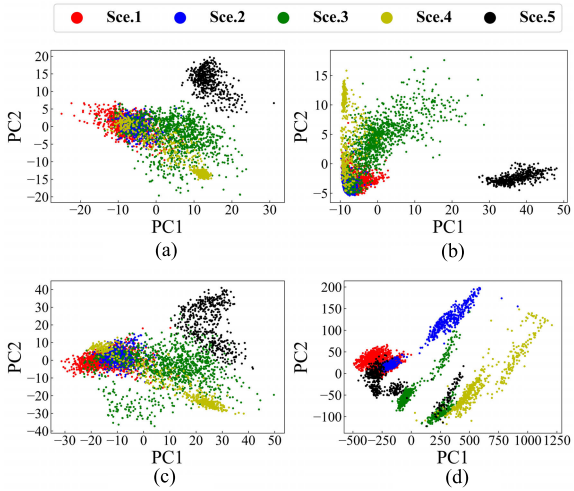


Fig. 14. Feature distribution map of different methods. (a) CNN. (b) ResNet. (c) MIPL-A. (d) MIPL-B.

The feature distribution maps of CNN and ResNet in the last convolution layer are shown in Fig. 14(a) and (b). We can see that both methods cannot separate the features of Sce.1, Sce.2, Sce.3 and Sce.4 correctly, which accounts for the poor overall performance of multiscenario classification. The features of MIPL-A and MIPL-B from the merge layer are shown in Fig. 14(c) and (d). Apparently, the proposed MIPL-B method can better separate scenarios with similar features. Besides, it is worth noting that the distribution of features from the time domain and visual domain is different. After feature fusion, the scenarios can be better separated. Although there are still several features close to each other, it does not affect the overall performance.

Furthermore, the confusion matrix contains detailed classification information selected to investigate the performance of each method. As shown in Fig. 15(a) and (c), both CNN and ResNet models can achieve scenario multiclassification to a certain extent. However, there are some ambiguous scenarios, such as Sce.2 and Sce.4, which cannot be accurately identified, leading to poor overall classification results. For the LSTM [see Fig. 15(b)] and SVM [see Fig. 15(d)], the misclassification is more pronounced, or sometimes they even fail to identify some scenarios (Sce.1, Sce.2 and Sce.4). For the proposed method, from Fig. 15(e) and (f) we can see that both MIPL-A and MIPL-B exhibit good performance in scenario multiclassification. Both MIPL-A and MIPL-B can identify Sce.5 with an accuracy of 100%. Compared to other traditional methods, the classification results of MIPL-A and MIPL-B are apparently improved, particularly for the Sce.1, Sce.2 and Sce.4. Additionally, it is evident that the inclusion of RSS features significantly benefits MIPL-B in identifying each scenario compared to MIPL-A. The identification precision for Sce.1 (95.55%), Sce.2 (88.49%), Sce.3 (98.11%) and Sce.4 (98.86%) is notably improved.

3) Performance Evaluation for Range Error Mitigation: In order to evaluate the NLOS mitigation performance of the proposed method and to investigate the advantages of NLOS subdivision. In the following experiments, the performance of the MIPL method for range error mitigation is compared with two commonly used methods including CNN and LSTM.

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

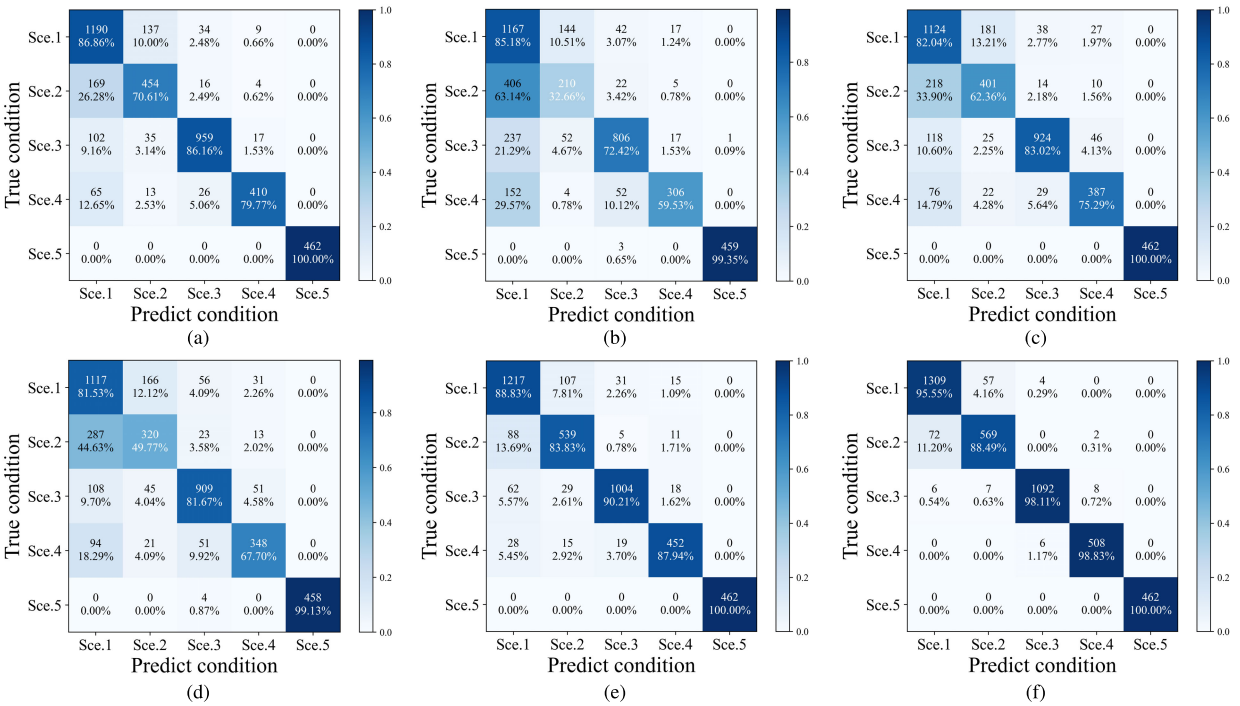


Fig. 15. Confusion matrix results of multiclassification. The Sce.1, Sce.2, Sce.3, Sce.4 and Sce.5 correspond to LOS, glass window, metal plate, corner, and door, respectively. (a) CNN. (b) LSTM. (c) ResNet. (d) SVM. (e) MIPL-A. (f) MIPL-B.

TABLE II
PERFORMANCE OF DIFFERENT METHODS FOR ERROR MITIGATION WITH UNCLASSIFIED DATA

Method	MAE(m)	RMSE(m)
Origin	0.6561	0.9682
CNN	0.2292	0.3502
stacked-LSTM	0.1991	0.2962
MIPL-A	0.1670	0.2629
MIPL-B	0.1359	0.2336

The dataset consisting of CIR signals is divided into three parts. The first part is CIR signals without classification. The second part is CIR signals with traditional LOS and NLOS classification. The third part is CIR signals with LOS and NLOS subdivisions.

Table II shows the comparison of different methods in range error mitigation. In this experiment, the unclassified CIR signals with their corresponding range error data are trained with different methods. The dataset has a starting MAE of 0.6561 m and a starting RMSE of 0.9682 m. From Table II we can see that the MIPL-B method has the lowest MAE (0.1359 m) and RMSE (0.2336 m). Compared with CNN, LSTM and MIPL-A the MAE of the proposed method is reduced by 40.71%, 31.74% and 18.63%; the RMSE is reduced by 33.29%, 21.34%, and 11.14%, respectively. Besides, compared with the starting MAE and RMSE, the proposed method reduced the MAE by 79.28% and the RMSE by 75.87%. Further, Fig. 16 shows the cumulative distribution function (cdf) curves of original range errors and mitigated range errors. As shown, the cdf performance of the proposed MIPL-B method is the best, with all range errors less than 0.2 m and over 80% range errors less than 0.1 m.

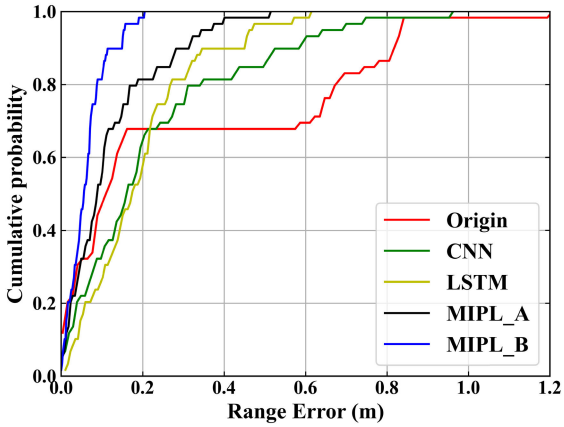


Fig. 16. CDF of four ranging mitigation algorithm with unlabeled data.

TABLE III
MAE AND RMSE OF CORRECTED RANGE ERROR USING BINARY CLASSIFIED DATA

Scenario	MAE(m)		RMSE(m)	
	Original	Mitigate	Original	Mitigate
LOS	0.0435	0.0287	0.0521	0.0305
NLOS	0.8093	0.1042	1.0822	0.1960
Overall	0.4264	0.0664	0.5672	0.1133

In the following experiment, we use the classified data to evaluate the error mitigation performance of MIPL-B in

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
MAE AND RMSE OF CORRECTED RANGE ERROR USING SUBDIVIDED DATA

Scenario	MAE(m)		RMSE(m)	
	Original	Mitigate	Original	Mitigate
LOS	0.0403	0.0249	0.0493	0.0274
NLOS #1	0.1250	0.0168	0.1189	0.0212
NLOS #2	0.7452	0.0276	0.7523	0.0356
NLOS #3	1.9001	0.1123	1.9701	0.2243
NLOS #4	0.4668	0.0279	0.4697	0.0387
Overall	0.6556	0.0419	0.6739	0.0695

different scenarios. In this case, we first use the MIPL-B method to train data classified as LOS and NLOS. The MAE and RMSE of mitigated range error from LOS and NLOS scenarios are shown in Table III. We can see that the overall MAE and RMSE reduced from 0.1359 and 0.2336 in Table II to 0.0664 and 0.1133, respectively. Further, we trained the model with NLOS data which is subdivided into four different scenarios to mitigate the range error. Table IV shows the MAE and RMSE of range error from four subdivided NLOS scenarios and LOS. The NLOS #1, NLOS #2, NLOS #3 and NLOS #4 correspond to glass, metal plate, corner, and door, respectively. Obviously, compared with LOS and NLOS binary classification, in Table IV the MAE and RMSE of each subdivided NLOS scenario are apparently lower, and the overall MAE and RMSE reduced by 36.89% and 38.66%. Besides, compared with the origin MAE and RMSE, the corrected MAE and RMSE are reduced by 93.61% and 89.69%.

Further, Fig. 17 shows the comparison of cdfs in five scenarios, from which we can see that the proposed MIPL method with NLOS subdivision can effectively mitigate NLOS. In each NLOS scenario, the range error is reduced by over 50%. Compared with the traditional NLOS mitigation method without NLOS subdivision, the proposed method has a more robust performance and can better mitigate the range error. It is worth noting that in LOS and NLOS #1 scenarios, the traditional NLOS mitigation method may increase the range error. As mentioned in the former article, the distribution features of NLOS range error are quite different in different scenarios. If we use the data without subdivision, the neural network may learn false information, which accounts for the inaccurate error prediction results. Although in the LOS scenario, the range errors are similar before and after error mitigation using the proposed method, the range errors are small enough, and will not affect the final localization results. All these experiments indicate that subdividing signal propagation scenarios can apparently improve the performance of error mitigation.

4) Indoor Localization Performance: In order to evaluate the performance of our NLOS identification and mitigation method in practical UWB-based indoor localization, we chose two indoor scenarios that contain all the NLOS scenarios mentioned in the previous article to carry out localization experiments. As shown in Fig. 18, there are four fixed anchors and a mobile tag in each scenario. The real position of each

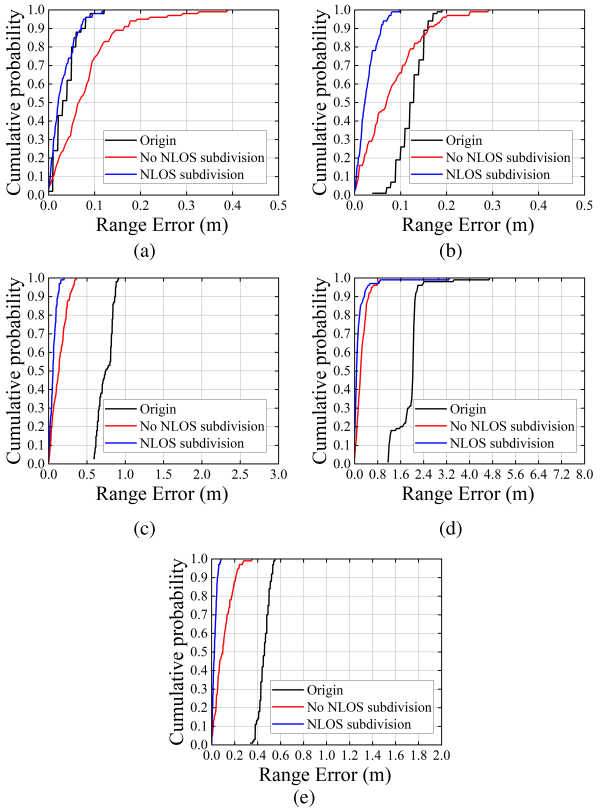


Fig. 17. CDFs of original range error and mitigated range error in five scenarios. (a) LOS. (b) NLOS #1. (c) NLOS #2. (d) NLOS #3. (e) NLOS #4.

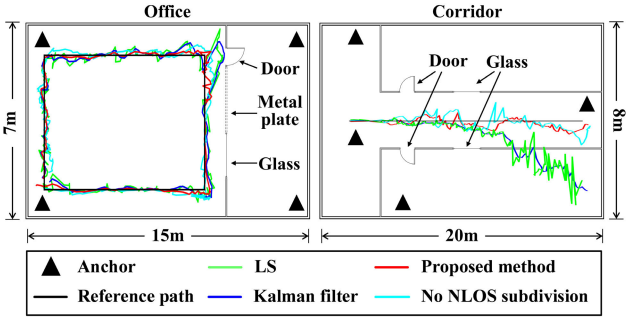


Fig. 18. Dynamic positioning experiments in two indoor scenarios.

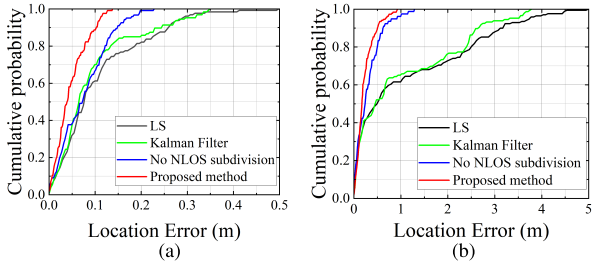


Fig. 19. CDFs of localization errors from different scenarios. (a) Office. (b) Corridor.

anchor is measured by a laser rangefinder. The tag moves along a reference path. The localization performance of the proposed method is compared with the least squares (LS) method, Kalman filter (KF) and traditional NLOS mitigation method without NLOS scenarios subdivision. Clearly, we can see that

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

TABLE V
INDOOR LOCALIZATION PERFORMANCE IN DIFFERENT SCENARIOS

Scenarios	Methods	MAE(m)	RMSE(m)
Office	LS	0.253	0.274
	LS+KF	0.174	0.203
	MIPL-B	0.068	0.081
	no NLOS subdivision	0.114	0.132
Corridor	LS	2.48	2.89
	LS+KF	1.88	2.19
	MIPL-B	0.461	0.535
	no NLOS subdivision	0.641	0.743

TABLE VI
TIME CONSUMPTION OF MIPL-B FOR THREE PHASES

Phase	Parameters	Time
Model training	12500	557 s
Model testing	1000	134 s
Data collection and preparation	15000	1496 s

our method can provide better localization results than other methods. Besides, compared with traditional NLOS mitigation methods without NLOS subdivision, our method can better alleviate the precision fluctuation caused by NLOS scenario switching. Since it is difficult to obtain the true position of the moving tag, we use cross-track error to further evaluate the localization accuracy. Fig. 19 shows the cdfs of localization errors from two indoor scenarios, from which we can also see that the performance of the proposed MIPL method is better than other methods. The location error of the proposed method from the office is less than 0.15 m and over 90% is less than 0.1 m. From the corridor, the location error is less than 1 m and over 90% is no more than 0.5 m. Additionally, Table V shows the MAE and RMSE of location error, from which we can see that the proposed MIPL method has the least MAE (0.068 m from office and 0.461 m from corridor) and RMSE (0.081 m from office and 0.535 m from corridor). Compared with the traditional method without NLOS subdivision with MAE 0.114 m from the office and 0.641 m from the corridor and RMSE 0.132 m from the office and 0.743 m from the corridor, the proposed MIPL method shows more advantages in error mitigation. Moreover, compared with the commonly used KF method, the performance of the MIPL method is also superior, especially in harsh NLOS scenarios (corridor).

Table VI shows the time consumption of the MIPL-B model for different phases. As shown, the most time-consuming phase is data collection and preparation. In this phase, 15 000 CIR data are collected from real indoor environments using UWB-based sensors. Subsequently, the CIR data is transformed into 2-D images. It is worth noting that the total time-consuming for the entire phase is less than 30 min, which is acceptable for practical use. The total time-consuming of the model training phase is 557 s, during which 12 500 data are used to train the NLOS identification and mitigation model. For the last model testing phase, 1000 CIR data is collected from real indoor localization experiments. We calculated the total time of data collection and transformation, scenario identification and range error prediction. We can see that the proposed method can complete the final NLOS mitigation task in 134 s, which can meet the requirements of indoor localization.

VI. CONCLUSION

In this article, we proposed a two-step NLOS identification and mitigation method based on the MIPL model and GAF encoding images. Our approach can fully leverage the potential of UWB CIR signals to identify and mitigate NLOS propagation, which requires no prior knowledge and can save much labor. Additionally, the UWB RSS information is considered as an auxiliary feature to further improve model NLOS identification and mitigation ability. Based on the LOS and NLOS identified results, the appropriate NLOS mitigation model is selected to correct NLOS range bias. In the NLOS identification stage, the proposed method is tested in a real indoor environment. Compared with traditional machine learning methods, the MIPL-A and MIPL-B methods both have good performance in traditional LOS/NLOS binary classification with accuracy of 91.35% and 97.36%, respectively. Then we subdivided the NLOS scenarios based on common indoor obstacles, and the multiclassification capability of the proposed method was tested. The results show that both MIPL-A and MIPL-B methods also have good performance on multiclassification, with overall accuracy of 90.18% and 96.24%, respectively. In the NLOS mitigation stage, the MIPL-B model is selected to predict NLOS range error. Using traditional NLOS data without subdivision, the MIPL-B method reduces the MAE and RMSE of range error from the original 0.6551 and 0.9682 m to 0.1359 and 0.2336 m, respectively. Moreover, using the subdivided NLOS data, the MIPL-B method can further reduce the MAE and RMSE of range error from the original 0.6556 and 0.6739 m to 0.0419 and 0.0695 m, respectively. In the indoor localization experiment, the proposed method can improve the localization accuracy by over 80%. In future work, we will further investigate how to apply the trained model in different indoor scenarios.

REFERENCES

[1] C. T. Li, J. C. P. Cheng, and K. Chen, "Top 10 technologies for indoor positioning on construction sites," *Autom. Construct.*, vol. 118, Oct. 2020, Art. no. 103309.
[2] Y. Lu and L. D. Xu, "Internet of Things (IoT) cybersecurity research: A review of current research topics," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 2103–2115, Apr. 2019.
[3] H.-C. Lee and K.-H. Ke, "Monitoring of large-area IoT sensors using a LoRa wireless mesh network system: Design and evaluation," *IEEE Trans. Instrum. Meas.*, vol. 67, no. 9, pp. 2177–2187, Sep. 2018.
[4] A. Alarifi et al., "Ultra wideband indoor positioning technologies: Analysis and recent advances," *Sensors*, vol. 16, no. 5, p. 707, May 2016.
[5] F. Zafari, A. Gkelias, and K. K. Leung, "A survey of indoor localization systems and technologies," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2568–2599, 3rd Quart., 2019.
[6] X. Zhu, J. Yi, J. Cheng, and L. He, "Adapted error map based mobile robot UWB indoor positioning," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 9, pp. 6336–6350, Sep. 2020.
[7] X. Qi and C. Chen, "A review of non-line-of-sight identification and mitigation algorithms forindoor localization," *Control Decis.*, vol. 37, no. 8, pp. 1921–1933, 2022.
[8] Y. Zou and H. Liu, "An efficient NLOS errors mitigation algorithm for TOA-based localization," *Sensors*, vol. 20, no. 5, p. 1403, Mar. 2020.

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.

[9] K. Yu, K. Wen, Y. Li, S. Zhang, and K. Zhang, "A novel NLOS mitigation algorithm for UWB localization in harsh indoor environments," *IEEE Trans. Veh. Technol.*, vol. 68, no. 1, pp. 686–699, Jan. 2019.

[10] L. Gazzah, L. Najjar, and H. Besbes, "Hybrid RSS/AOA hypothesis test for NLOS/LOS base station discrimination and location error mitigation," *Trans. Emerg. Telecommun. Technol.*, vol. 27, no. 5, pp. 626–639, May 2016.

[11] J. Wang, J. Wen, J. Hua, X. Feng, Z. Xu, and Z. Ni, "Angle residual-based NLOS suppression and localization method in wireless sensor networks," *IEEE Sensors J.*, vol. 23, no. 13, pp. 14388–14396, Jul. 2023.

[12] Y. Dong, T. Arslan, and Y. Yang, "Real-time NLOS/LOS identification for smartphone-based indoor positioning systems using WiFi RTT and RSS," *IEEE Sensors J.*, vol. 22, no. 6, pp. 5199–5209, Mar. 2022.

[13] F. Che et al., "Feature-based generalized Gaussian distribution method for NLoS detection in ultra-wideband (UWB) indoor positioning system," *IEEE Sensors J.*, vol. 22, no. 19, pp. 18726–18739, Oct. 2022.

[14] X. Yang, "NLOS mitigation for UWB localization based on sparse pseudo-input Gaussian process," *IEEE Sensors J.*, vol. 18, no. 10, pp. 4311–4316, May 2018.

[15] B. J. Silva and G. P. Hancke, "Non-line-of-sight identification without channel statistics," in *Proc. 46th Annu. Conf. IEEE Ind. Electron. Soc. (IECON)*, Oct. 2020, pp. 4489–4493.

[16] K. K. Garg, P. Shaik, V. Bhatia, and O. Krejcar, "On the performance of a relay assisted hybrid RF-NLOS UVC system with imperfect channel estimation," *J. Opt. Commun. Netw.*, vol. 14, no. 4, pp. 177–189, Apr. 2022.

[17] Z. Zeng, R. Bai, L. Wang, and S. Liu, "NLOS identification and mitigation based on CIR with particle filter," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Apr. 2019, pp. 1–6.

[18] Y. Wang, K. Gu, Y. Wu, W. Dai, and Y. Shen, "NLOS effect mitigation via spatial geometry exploitation in cooperative localization," *IEEE Trans. Wireless Commun.*, vol. 19, no. 9, pp. 6037–6049, Sep. 2020.

[19] X. Yang, J. Wang, D. Song, B. Feng, and H. Ye, "A novel NLOS error compensation method based IMU for UWB indoor positioning system," *IEEE Sensors J.*, vol. 21, no. 9, pp. 11203–11212, May 2021.

[20] J. B. Kristensen, M. Massanet Ginard, O. K. Jensen, and M. Shen, "Non-line-of-sight identification for UWB indoor positioning systems using support vector machines," in *IEEE MTT-S Int. Microw. Symp. Dig.*, May 2019, pp. 1–3.

[21] H. Yang, Y. Wang, C. K. Seow, M. Sun, M. Si, and L. Huang, "UWB sensor-based indoor LOS/NLOS localization with support vector machine learning," *IEEE Sensors J.*, vol. 23, no. 3, pp. 2988–3004, Feb. 2023.

[22] C. Huang et al., "Machine learning-enabled LOS/NLOS identification for MIMO systems in dynamic environments," *IEEE Trans. Wireless Commun.*, vol. 19, no. 6, pp. 3643–3657, Jun. 2020.

[23] Z. Xiao, H. Wen, A. Markham, N. Trigoni, P. Blunsom, and J. Frolik, "Non-line-of-sight identification and mitigation using received signal strength," *IEEE Trans. Wireless Commun.*, vol. 14, no. 3, pp. 1689–1702, Mar. 2015.

[24] T. Wang, K. Hu, Z. Li, K. Lin, J. Wang, and Y. Shen, "A semi-supervised learning approach for UWB ranging error mitigation," *IEEE Wireless Commun. Lett.*, vol. 10, no. 3, pp. 688–691, Mar. 2021.

[25] J. Fan and A. S. Awan, "Non-line-of-sight identification based on unsupervised machine learning in ultra wideband systems," *IEEE Access*, vol. 7, pp. 32464–32471, 2019.

[26] M. Stahlke, S. Kram, C. Mutschler, and T. Mahr, "NLOS detection using UWB channel impulse responses and convolutional neural networks," in *Proc. Int. Conf. Localization GNSS (ICL-GNSS)*, Jun. 2020, pp. 1–6.

[27] T. Chang, S. Jiang, Y. Sun, A. Jia, and W. Wang, "Multi-bandwidth NLOS identification based on deep learning method," in *Proc. 15th Eur. Conf. Antennas Propag. (EuCAP)*, Mar. 2021, pp. 1–5.

[28] C. Jiang, J. Shen, S. Chen, Y. Chen, D. Liu, and Y. Bo, "UWB NLOS/LOS classification using deep learning method," *IEEE Commun. Lett.*, vol. 24, no. 10, pp. 2226–2230, Oct. 2020.

[29] C. Jiang, S. Chen, Y. Chen, D. Liu, and Y. Bo, "An UWB channel impulse response de-noising method for NLOS/LOS classification boosting," *IEEE Commun. Lett.*, vol. 24, no. 11, pp. 2513–2517, Nov. 2020.

[30] Q. Liu, Z. Yin, Y. Zhao, Z. Wu, and M. Wu, "UWB LOS/NLOS identification in multiple indoor environments using deep learning methods," *Phys. Commun.*, vol. 52, Jun. 2022, Art. no. 101695.

[31] Y. Pei, R. Chen, D. Li, X. Xiao, and X. Zheng, "FCN-attention: A deep learning UWB NLOS/LOS classification algorithm using fully convolution neural network with self-attention mechanism," *Geo-Spatial Inf. Sci.*, pp. 1–20, Mar. 2023, doi: 10.1080/10095020.2023.2178334.

[32] J. Wei, H. Wang, S. Su, Y. Tang, X. Guo, and X. Sun, "NLOS identification using parallel deep learning model and time-frequency information in UWB-based positioning system," *Measurement*, vol. 195, May 2022, Art. no. 111191.

[33] J. Wang, K. Yu, J. Bu, Y. Lin, and S. Han, "Multi-classification of UWB signal propagation channels based on one-dimensional wavelet packet analysis and CNN," *IEEE Trans. Veh. Technol.*, vol. 71, no. 8, pp. 8534–8547, Aug. 2022.

[34] H. Tang, Z. Liao, P. Chen, D. Zuo, and S. Yi, "A novel convolutional neural network for low-speed structural fault diagnosis under different operating condition and its understanding via visualization," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–11, 2021.

[35] Y. Zhou, X. Long, M. Sun, and Z. Chen, "Bearing fault diagnosis based on Gramian angular field and DenseNet," *Math. Biosci. Eng.*, vol. 19, no. 12, pp. 14086–14101, 2022.

[36] C. Camara, P. Peris-Lopez, M. Safkhani, and N. Bagheri, "ECG identification based on the Gramian angular field and tested with individuals in resting and activity states," *Sensors*, vol. 23, no. 2, p. 937, Jan. 2023.

[37] X. Yang, Y. Ye, X. Li, R. Y. K. Lau, X. Zhang, and X. Huang, "Hyperspectral image classification with deep learning models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5408–5423, Sep. 2018.



Bowen Deng (Graduate Student Member, IEEE) was born in Jinan, Shandong, China, in 1998. He received the B.S. degree from Chang'an University, Xi'an, China, in 2021, where he is currently pursuing the M.S. degree in electronic and control engineering.

His research interests include indoor positioning and machine learning.



Tangwen Xu was born in Xi'an, Shaanxi, China, in 1984. She received the B.S. and Ph.D. degrees from the School of Telecommunication Engineering, Xidian University, Xi'an, in 2006 and 2014, respectively. She is pursuing the combined master's and doctoral program.

She is currently a Lecturer with the School of Electronic and Control Engineering, Chang'an University, Xi'an. Her research interests include collaborative communication, cognitive radio, spectrum sharing, and energy acquisition technology.



Maode Yan received the B.S., M.S., and Ph.D. degrees from the School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an, China, in 1996, 1999, and 2001, respectively.

He is currently a Professor with the School of Electronic and Control Engineering, Chang'an University, Xi'an. His research interests include networked control systems, vehicle platoon modeling and control, robots and their formation control, and embedded systems and applications.

Authorized licensed use limited to: Chang Gung Univ.. Downloaded on December 24,2025 at 09:33:08 UTC from IEEE Xplore. Restrictions apply.